

Name: Solution

MATH 241

QUIZ#5

SUMMER 2014

Problem 1. (5 Points)

Consider the linear system of equations in the matrix form $A\mathbf{x}=\mathbf{b}$ and let the reduced row echelon form of the augmented matrix $[A|\mathbf{b}]$ is obtained by Gauss-Jordan elimination as $[R|\mathbf{v}]$ where

$R = \begin{bmatrix} I_{r \times r} & N_{r \times f} \\ 0_{k \times r} & 0_{k \times f} \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} \mathbf{c}_{r \times 1} \\ 0_{k \times 1} \end{bmatrix}$. $I_{r \times r}$ is the r by r identity matrix, $N_{r \times f}$ is an r by f matrix, $\mathbf{c}_{r \times 1} \in \mathbb{R}^r$ and $0_{m \times n}$ is the m by n zero matrix.

- 2 (a) What is the rank of A ? Is the linear system consistent? Please explain your answer!
- 2 (b) Assume that we have the sequence of free variables in the vector form.

$$\text{i.e. } [x_{r+1} \ x_{r+2} \ \dots \ x_{r+f}]^T = [\alpha_1 \ \dots \ \alpha_f]^T = \alpha.$$

Find the solution $\mathbf{x} = [x_1 \ \dots \ x_{r+f}]^T$ in terms of $I_{r \times r}$, $N_{r \times f}$, $\mathbf{c}_{r \times 1}$, α and appropriate size of zero and identity matrices. Verify that $R\mathbf{x}=\mathbf{v}$.

- 1 (c) Find a basis for $\text{Row}(A)$, $\text{Null}(A)$ in terms of $I_{r \times r}$, $N_{r \times f}$ and an appropriate size of identity matrix. Which columns of A certainly form a basis for $\text{Col}(A)$?

(a)

$$\begin{array}{ccccccc} \textcircled{p} & & \textcircled{p} & \textcircled{p} & \dots & \textcircled{p} & \\ \downarrow & & \downarrow & \downarrow & & & \\ \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & * & \dots & * \\ 0 & 1 & 0 & \dots & 0 & * & \dots & * \\ 0 & 0 & 1 & \dots & 0 & * & \dots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & * & \dots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{bmatrix} \end{array}$$

if we observe the expanded R matrix on the left, we find out that $I_{r \times r}$ contains the pivots and those columns are the pivot columns. Therefore we have r pivots.

$$\# \text{ pivots} = \# \text{ LI rows (columns)} = \text{rank}(A) = \text{rank}(R)$$

We have consistent system, since for the zero rows of R we have zero coefficients on the right hand side!

$$\text{or } \text{rank}([A|\mathbf{b}]) = \text{rank}([R|\mathbf{v}]) = \text{rank}(R) = \text{rank}(A)$$

$$\begin{cases} x_1 = c_1 - \alpha_1 N_{11} - \alpha_2 N_{12} - \dots - \alpha_f N_{1f} \\ x_2 = c_2 - \alpha_1 N_{21} - \alpha_2 N_{22} - \dots - \alpha_f N_{2f} \\ \vdots \\ x_r = c_r - \alpha_1 N_{r1} - \alpha_2 N_{r2} - \dots - \alpha_f N_{rf} \\ x_{r+1} = 0 + \alpha_1 \cdot 1 + \alpha_2 \cdot 0 - \dots + \alpha_f \cdot 0 \\ x_{r+2} = 0 + \alpha_1 \cdot 0 + \alpha_2 \cdot 1 + \dots + \alpha_f \cdot 0 \\ \vdots \\ x_{r+f} = 0 + \alpha_1 \cdot 0 + \alpha_2 \cdot 0 + \dots + \alpha_f \cdot 1 \end{cases}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{r+f} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_r \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} -N_{11} & \dots & -N_{1f} \\ -N_{21} & \dots & -N_{2f} \\ \vdots & \ddots & \vdots \\ -N_{r1} & \dots & -N_{rf} \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_f \end{bmatrix} = \begin{bmatrix} \mathbf{c}_{r \times 1} \\ 0_{f \times 1} \end{bmatrix} + \begin{bmatrix} -N_{r \times f} \\ I_{f \times f} \end{bmatrix} \alpha_{f \times 1}$$

$$R \cdot \mathbf{x} = \begin{bmatrix} I_r \\ 0 \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ 0 \end{bmatrix} + \begin{bmatrix} -N \\ I_f \end{bmatrix} \alpha = \begin{bmatrix} I_r \cdot \mathbf{c} + 0 \\ 0 \cdot \mathbf{c} + 0 \cdot 0 \end{bmatrix} + \begin{bmatrix} -N \\ I_f \end{bmatrix} \alpha = \begin{bmatrix} \mathbf{c} \\ 0 \end{bmatrix} = \mathbf{v}$$

- (c) $\mathbf{u} \in \text{Row}(A)$ $\mathbf{u} = [I_{r \times r} \ N_{r \times f}]^T \cdot \alpha$ $\alpha \in \mathbb{R}^f$ non zero rows of R spans $\text{Row}(A)$
 - $\mathbf{n} \in \text{Null}(A)$ $\mathbf{n} = \begin{bmatrix} -N_{r \times f} \\ I_{f \times f} \end{bmatrix} \alpha$ $\alpha \in \mathbb{R}^f$ Null space is the homogeneous solution.
- First r columns of A certainly independent since they are pivot columns.

(Another way)!

(b) let $\underline{x} = \begin{bmatrix} \underline{x}_{rx1} \\ \underline{x}_{fx1} \end{bmatrix}$, $\underline{x}_{fx1} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_f \end{bmatrix} = \underline{\alpha}$ is given

$$\underline{R} \underline{x} = \begin{bmatrix} \underline{I}_r & \underline{N} \\ \underline{0} & \underline{0} \end{bmatrix} \begin{bmatrix} \underline{x}_r \\ \underline{x}_f \end{bmatrix} = \begin{bmatrix} \underline{I}_r \underline{x}_r + \underline{N} \underline{x}_f \\ \underline{0} \end{bmatrix} = \begin{bmatrix} \underline{c}_r \\ \underline{0} \end{bmatrix}$$

$$\underbrace{\underline{I}_r}_{r \times r} \underline{x}_r + \underbrace{\underline{N}}_{r \times f} \underbrace{\underline{x}_f}_{f \times 1} = \underline{c}_r \Rightarrow \left. \begin{array}{l} \underline{x}_r = \underline{c}_r - \underline{N} \underline{\alpha} \\ \underline{x}_f = \underline{I} \cdot \underline{\alpha} \end{array} \right\} \Rightarrow \begin{bmatrix} \underline{x}_r \\ \underline{x}_f \end{bmatrix} = \begin{bmatrix} \underline{c}_r \\ \underline{0} \end{bmatrix} + \begin{bmatrix} -\underline{N} \\ \underline{I} \end{bmatrix} \underline{\alpha}$$

$$\begin{bmatrix} \underline{I}_r & \underline{N} \\ \underline{0} & \underline{0} \end{bmatrix} \left(\begin{bmatrix} \underline{c}_r \\ \underline{0} \end{bmatrix} + \begin{bmatrix} -\underline{N} \\ \underline{I} \end{bmatrix} \underline{\alpha} \right) = \begin{bmatrix} \underline{I}_r \underline{c}_r + \underline{N} \underline{0} \\ \underline{0} \end{bmatrix} + \underbrace{\begin{bmatrix} -\underline{I}_r \underline{N} + \underline{N} \underline{I} \\ -\underline{0} \underline{N} + \underline{0} \underline{I} \end{bmatrix}}_{\underline{0}} \underline{\alpha} = \begin{bmatrix} \underline{c}_r \\ \underline{0} \end{bmatrix}$$

Problem 2. (5 Points)

- 3 (a) We define the determinant of A as $\det(A) = \sum_{k=1}^n a_{kj} A_{kj} = \sum_{k=1}^n a_{ik} A_{ik}$, for any column j or any row i where the cofactor $A_{ij} = (-1)^{(i+j)} M_{ij}$ and the minor M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix that survives when the i^{th} row and the j^{th} column of A are removed. Given the determinant definition above, prove that $\det(A^T) = \det(A)$ by using the method of induction.

- 2 (b) Consider an upper triangular square matrix U with nonzero diagonal entries. Show that the columns of U is linearly independent. i.e. $\sum_{j=1}^n \alpha_j \mathbf{u}_j = \mathbf{0}$ is satisfied only for all α_j 's are zero where $\mathbf{u}_j$'s are columns of U .

base case: A is 1×1 matrix $A_{11} = [a_{11}] \det(A^T) = \det(A) \Rightarrow A^T = A$

(a) $\det(A^T) = \sum_{k=1}^n a_{kj}^T A_{kj}^T = \sum_{k=1}^n a_{jk} A_{kj}^T \rightarrow \text{Defn of transpose}$

$A^T = \begin{bmatrix} \underline{B} & \underline{C} \\ \underline{D} & \underline{E} \end{bmatrix} \xrightarrow{\text{row } k^{\text{th}} \text{ column removed}} \sum_{k=1}^n a_{jk} (-1)^{k+j} M_{kj}^T \rightarrow \text{Defn of cofactor}$

$\Rightarrow \sum_{k=1}^n a_{jk} (-1)^{k+j} M_{jk} \rightarrow \text{induction hypothesis}$

$M_{jk} = |S_{jk}|$

$A^T = \begin{bmatrix} \underline{B}^T & \underline{D}^T \\ \underline{C}^T & \underline{E}^T \end{bmatrix} \Rightarrow S_{kj}^T = \begin{bmatrix} \underline{B}^T & \underline{D}^T \\ \underline{C}^T & \underline{E}^T \end{bmatrix} = (S_{jk})^T = \sum_{k=1}^n a_{jk} A_{jk} = \det(A)$

$M_{kj}^T = |S_{kj}^T| = |(S_{jk})^T| = |S_{jk}| = M_{jk}$

$\hookrightarrow$ by induction hypothesis, since S_{jk} is $(n-1) \times (n-1)$ matrix

- (b) $\sum_{j=1}^n \alpha_j \mathbf{u}_j = \mathbf{0} \Rightarrow \alpha = \mathbf{0} \rightarrow$ for column n $0\alpha_1 + \alpha_2 \cdot 0 + \dots + \alpha_n \cdot u_{nn} = 0 \Rightarrow \alpha_n = 0$
- for column $n-1$ $0\alpha_1 + 0\alpha_2 + \dots + \alpha_{n-1} \cdot u_{(n-1)(n-1)} + 0 \cdot \alpha_n = 0 \Rightarrow \alpha_{n-1} = 0$
- we go on like this
- for column 2 $\alpha_1 \cdot 0 + \alpha_2 u_{22} + 0 \cdot u_{23} + \dots + 0 \cdot u_{2n} = 0 \Rightarrow \alpha_2 = 0$
- for column 1 $\alpha_1 u_{11} + 0 \cdot u_{12} + 0 \cdot u_{13} + \dots + 0 \cdot u_{1n} = 0 \Rightarrow \alpha_1 = 0$
- so all α_i 's are zero $\Rightarrow \mathbf{u}_j$'s are linearly independent